

The Silent Tax: Typo-Robustness Analysis (GSM8K)

DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named `typo{rate}_real{ratio}`: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config. `n` = questions; `n_answered` = questions that produced a final answer (not truncated at the token cap). `acc_answered` = correct / `n_answered`; `acc_strict` = correct / `n` (truncated counted as wrong); `ci_lo/ci_hi` = 95% bootstrap CI on `acc_strict`.

config	n	n_answered	acc_answered	acc_strict	ci_lo	ci_hi
clean	500	481	0.9522	0.916	0.89	0.94
typo25_real10	500	454	0.9075	0.824	0.792	0.856
typo25_real40	500	458	0.9083	0.832	0.8	0.864
typo25_real70	500	454	0.8943	0.812	0.776	0.848
typo50_real10	500	446	0.843	0.752	0.714	0.788
typo50_real40	500	445	0.8247	0.734	0.696	0.772
typo50_real70	500	433	0.7898	0.684	0.642	0.724
typo75_real10	500	412	0.7816	0.644	0.602	0.686
typo75_real40	500	417	0.7626	0.636	0.594	0.68
typo75_real70	500	392	0.7143	0.56	0.514	0.604

Table 2: Answered-only flips vs clean, paired by question over `n_both` (answered in both conditions). `r2w` / `w2r` = questions correct on clean but wrong on this config / wrong on clean but correct on this config; `r2w_ratio` = `r2w` / `n_both`, `w2r_ratio` = `w2r` / `n_both`; `net_flip_ratio` = `r2w_ratio` - `w2r_ratio` = (`r2w` - `w2r`) / `n_both`, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
typo25_real10	448	30	8	0.067	0.0179	0.0491	30	8	11.605	0.00065765
typo25_real40	452	27	9	0.0597	0.0199	0.0398	27	9	8.028	0.00460653
typo25_real70	448	33	6	0.0737	0.0134	0.0603	33	6	17.333	3.136e-05
typo50_real10	438	53	4	0.121	0.0091	0.1119	53	4	40.421	0.0
typo50_real40	440	62	4	0.1409	0.0091	0.1318	62	4	49.227	0.0
typo50_real70	428	78	8	0.1822	0.0187	0.1636	78	8	55.36	0.0
typo75_real10	409	75	4	0.1834	0.0098	0.1736	75	4	62.025	0.0
typo75_real40	412	84	6	0.2039	0.0146	0.1893	84	6	65.878	0.0
typo75_real70	383	97	3	0.2533	0.0078	0.2454	97	3	86.49	0.0

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: *b* = right→wrong, *c* = wrong→right) and ignores those that stayed the same.

- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column `mcnemar_stat`); `mcnemar_p` is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when `mcnemar_p` < 0.05 — the right→wrong vs wrong→right imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	481	1388.5	1200	2315	4096
typo25_real10	454	1348.2	1140.5	2494	4096
typo25_real40	458	1418.9	1219.0	2670	4096
typo25_real70	454	1372.2	1164.5	2350	4096
typo50_real10	446	1464.1	1226.5	2820	4096
typo50_real40	445	1466.4	1203	2896	4096
typo50_real70	433	1473.4	1227	2798	4096
typo75_real10	412	1539.3	1254.5	3012	4096
typo75_real40	417	1571.2	1395	2997	4096
typo75_real70	392	1588.9	1290.5	3110	4096

Table 4: Mean generated tokens for correct vs wrong answers. `wrong_over_correct` = `tok_wrong` / `tok_correct` (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	1333.6	2480.5	1.86
typo25_real10	1278.6	2031.4	1.5888
typo25_real40	1379.3	1810.7	1.3128
typo25_real70	1301.4	1970.5	1.5141
typo50_real10	1379.3	1919.6	1.3917
typo50_real40	1392.4	1814.5	1.3031
typo50_real70	1347.7	1945.8	1.4438
typo75_real10	1393.1	2062.3	1.4803
typo75_real40	1454.4	1946.5	1.3384
typo75_real70	1401.0	2058.8	1.4695

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check*.
- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused*.

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait*, *hmm*, *actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo*, *misspelled*, *doesn't make sense*, ...) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n.answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	481	12.924	9.757	5.783	4.366	18.707	14.123
typo25_real10	454	13.266	9.731	8.455	6.203	21.721	15.934
typo25_real40	458	13.196	10.284	9.24	7.201	22.436	17.485
typo25_real70	454	13.056	9.795	8.546	6.412	21.602	16.207
typo50_real10	446	13.385	10.87	10.878	8.834	24.263	19.704
typo50_real40	445	13.333	10.739	11.087	8.93	24.42	19.67
typo50_real70	433	13.639	11.189	13.465	11.046	27.104	22.236
typo75_real10	412	13.695	11.682	15.38	13.119	29.075	24.801
typo75_real40	417	13.629	11.914	15.121	13.218	28.751	25.132
typo75_real70	392	13.162	11.801	16.565	14.852	29.727	26.653

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	2.407	6.188	3.781
or maybe/alt	uncertainty	2.03	2.937	0.908
wait	second_guess	5.488	6.394	0.906
perhaps	uncertainty	0.432	1.182	0.749
unclear/confused	uncertainty	0.446	0.889	0.443
might be	uncertainty	0.088	0.338	0.25
not sure	uncertainty	0.014	0.155	0.141
hmm	second_guess	3.109	3.233	0.124
but wait	second_guess	0.584	0.663	0.079
I assume	uncertainty	0.011	0.054	0.043
not entirely	uncertainty	0.0	0.032	0.032
could be	uncertainty	0.135	0.158	0.023
possibly	uncertainty	0.0	0.006	0.006
presumably	uncertainty	0.003	0.007	0.004
reconsider	second_guess	0.0	0.001	0.001
on second thought	second_guess	0.0	0.0	0.0
I guess	uncertainty	0.218	0.173	-0.045
actually	second_guess	0.405	0.345	-0.06
hold on	second_guess	0.843	0.779	-0.063
wait/actually no	second_guess	0.435	0.302	-0.133
let me recheck	second_guess	0.961	0.776	-0.185
double-check	second_guess	1.099	0.882	-0.217

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	12.618	15.863	1.257	5.377	9.688	1.802	17.996	25.551	1.42
typo25_real10	12.993	14.793	1.139	7.008	16.555	2.362	20.0	31.348	1.567
typo25_real40	12.886	15.392	1.194	8.564	14.032	1.638	21.451	29.424	1.372
typo25_real70	12.933	13.719	1.061	7.375	14.885	2.018	20.308	28.604	1.409
typo50_real10	13.158	14.21	1.08	9.32	16.557	1.777	22.479	30.767	1.369
typo50_real40	12.883	14.875	1.155	9.744	15.691	1.61	22.628	30.566	1.351
typo50_real70	13.133	14.85	1.131	11.177	18.938	1.694	24.31	33.788	1.39
typo75_real10	13.281	14.641	1.102	13.236	20.271	1.532	26.517	34.911	1.317
typo75_real40	13.141	14.719	1.12	12.537	20.89	1.666	25.678	35.609	1.387
typo75_real70	12.798	13.72	1.072	12.577	22.682	1.803	25.376	36.403	1.435

LLM judge. The proposal specifies a second, non-lexical measure for both dimensions: a separate model with a fixed, strict prompt. The judge is a different model from the one under test, called at temperature 0 on a seeded random subsample of answered traces per config. It is shown the corrupted-word list (*original* $\rightarrow$ *as shown*), the question, and the reasoning trace (head+tail window). It returns two scalar ratings and evidence: *self_doubt_score* from 0–10, where 0 means no second-guessing and 10 means pervasive uncertainty or looping; and *repair_understanding_score* from 0–5, where 0 means the question’s meaning is fully understood and 5 means the model is fundamentally lost because of the corruption. The judge also returns a one-line justification and verbatim evidence quotes; quotes are checked against the reasoning trace. The judge is a subsample measure, while the lexical tables remain the full-data result.

Table 8: New scalar LLM-judge analysis per config. mean/median *self_doubt_score* use the 0-10 scale; mean/median *repair_understanding_score* use the 0-5 scale; threshold columns show the share of traces with substantial doubt (≥ 5) or substantial loss of meaning (≥ 3).

config	n	mean_self_doubt	median_self_doubt	frac_self_doubt_ge5	mean_repair_understanding	median_repair_understanding	frac_repair_understanding_ge3	evidence_self_doubt_frac	evidence_repair_frac
clean	494	0.684	0	0.0607	0.053	0	0.0081	0.9939	0.998
typo25_real10	481	1.181	0	0.1206	0.47	0	0.0769	0.9813	0.9356
typo25_real40	484	1.054	0	0.1054	0.705	0	0.1322	0.9793	0.938
typo25_real70	485	0.924	0	0.0907	0.565	0	0.101	0.9629	0.934
typo50_real10	484	0.89	0	0.0868	0.531	0	0.0971	0.9814	0.936
typo50_real40	486	0.802	0	0.0864	0.572	0	0.1173	0.9856	0.9527
typo50_real70	484	0.843	0	0.0868	0.624	0	0.1178	0.9855	0.9566
typo75_real10	482	0.934	0	0.1058	0.66	0	0.1203	0.9793	0.9378
typo75_real40	487	0.754	0	0.0903	0.815	0	0.1499	0.9836	0.9466
typo75_real70	486	0.712	0	0.0823	0.817	0	0.1667	0.9753	0.9465

Table 9: New scalar LLM-judge scores split by final answer outcome. The correct/wrong columns are means over traces whose final answer was correct or wrong.

config	n	n_correct	n_wrong	mean_doubt_correct	mean_doubt_wrong	mean_repair_correct	mean_repair_wrong
clean	494	413	81	0.654	0.84	0.044	0.099
typo25_real10	481	371	110	1.127	1.364	0.353	0.864
typo25_real40	484	357	127	1.036	1.102	0.49	1.307
typo25_real70	485	369	116	0.916	0.948	0.393	1.112
typo50_real10	484	336	148	0.985	0.676	0.473	0.662
typo50_real40	486	324	162	0.765	1.056	0.377	0.963
typo50_real70	484	295	189	0.881	0.783	0.393	0.984
typo75_real10	482	303	179	0.776	1.201	0.36	1.168
typo75_real40	487	266	221	0.789	0.71	0.481	1.217
typo75_real70	486	233	253	0.451	0.953	0.451	1.154
all	4853	3267	1586	0.852	0.946	0.37	1.014

Table 10: New scalar LLM-judge scores pooled by real-word typo ratio, showing whether higher real-word corruption is associated with more doubt or loss of understanding.

real_ratio	n	mean_self_doubt	mean_repair_understanding	frac_self_doubt_ge5	frac_repair_understanding_ge3
10	1447	1.001	0.554	0.1044	0.0981
40	1457	0.889	0.697	0.094	0.1332
70	1455	0.826	0.669	0.0866	0.1285

Repair Behaviour

Table 11: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo25_real10	3307	2472	324	93	418	0.7475	0.098	0.0281	0.1264
typo25_real40	3316	2345	438	146	387	0.7072	0.1321	0.044	0.1167
typo25_real70	3288	2279	482	138	389	0.6931	0.1466	0.042	0.1183
typo50_real10	6299	4312	1006	276	705	0.6846	0.1597	0.0438	0.1119
typo50_real40	6271	4021	1148	283	819	0.6412	0.1831	0.0451	0.1306
typo50_real70	6089	3553	1376	412	748	0.5835	0.226	0.0677	0.1228
typo75_real10	8088	4788	1712	513	1075	0.592	0.2117	0.0634	0.1329
typo75_real40	8214	4571	2042	576	1025	0.5565	0.2486	0.0701	0.1248
typo75_real70	7615	3943	1978	663	1031	0.5178	0.2598	0.0871	0.1354

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* → *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns ***_n** are counts; ***_frac** are the share of **n_corrupt_words**.

Table 12: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	17694	0.654	0.1719	0.0498	0.1242
40	17801	0.6144	0.2038	0.0565	0.1253
70	16992	0.5753	0.2258	0.0714	0.1276

Table 13: Misread rate split by final correctness. A typo’d word is ‘misread’ when the reasoning uses only its corrupted form (e.g. sun instead of sum). Within each outcome group, misread rate = (typo’d words that were misread) / (all typo’d words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo25_real10	0.0267	0.0419	1.571
typo25_real40	0.0407	0.0762	1.874
typo25_real70	0.0409	0.0507	1.239
typo50_real10	0.0426	0.0497	1.166
typo50_real40	0.0421	0.0584	1.388
typo50_real70	0.0656	0.0744	1.135
typo75_real10	0.0573	0.0836	1.46
typo75_real40	0.061	0.0968	1.586
typo75_real70	0.0713	0.1204	1.689

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and p -values in the tables below for this dataset’s result.

Table 14: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. acc_low_real / acc_high_real = accuracy on the lower / higher real-word variant; delta_acc = acc_high_real - acc_low_real; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with r2w $\hat{>}$ w2r means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	425	0.9176	0.9129	-0.0047	21	19	0.87436706
25	real10_to_real40	429	0.9138	0.9161	0.0023	17	18	1.0
25	real40_to_real70	433	0.9169	0.9007	-0.0162	30	23	0.40984673
50	real10_to_real70	399	0.8521	0.802	-0.0501	52	32	0.03816577
50	real10_to_real40	411	0.8564	0.8418	-0.0146	37	31	0.54428962
50	real40_to_real70	411	0.8345	0.8078	-0.0268	39	28	0.2218232
75	real10_to_real70	346	0.8237	0.7341	-0.0896	56	25	0.00085812
75	real10_to_real40	366	0.8033	0.7978	-0.0055	44	42	0.91412835
75	real40_to_real70	354	0.7994	0.7514	-0.048	42	25	0.05061718

Table 15: Logistic regression correct $\sim$ num_real + num_nonword over all answered typo traces (num_real / num_nonword = count of real-word / non-word typos in the problem). coef_* = change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful); odds_* = exp(coef); intercept is the model constant. n = number of traces fit.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
3911	-0.06822	-0.03564	0.9341	0.965	2.34646

Table 16: Among WRONG answers only, does the model flag the corruption, by real ratio? explicit_notice_frac = fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn't make sense / ...); silent_frac = 1 - explicit_notice_frac. The proposal predicts real-word failures are quieter (explicit_notice_frac should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	202	0.5644	0.4356
40	219	0.5479	0.4521
70	251	0.6175	0.3825

Lexical Grid (overview)

Table 17: All four marker families (second_guess, uncertainty, typo_noticing, repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	12.924	5.783	1.418	1.151
typo25_real10	13.266	8.455	2.873	2.132
typo25_real40	13.196	9.24	3.082	2.269
typo25_real70	13.056	8.546	3.042	2.09
typo50_real10	13.385	10.878	3.512	2.419
typo50_real40	13.333	11.087	3.705	2.614
typo50_real70	13.639	13.465	4.577	3.403
typo75_real10	13.695	15.38	4.658	3.278
typo75_real40	13.629	15.121	4.365	3.053
typo75_real70	13.162	16.565	4.53	3.426

Spellcheck Recovery

Table 18: External spellcheck recovery by config. restored_exact_pct = percentage of typo words that were corrected back to exactly their original clean word.

config	n_questions	total_typo_words	restored_exact_n	changed_other_n	not_restored_n	restored_exact_frac	restored_exact_pct
typo25_real10	500	6500	4000	1000	2500	0.615385	61.54
typo25_real40	500	6500	3500	500	3000	0.538462	53.85
typo25_real70	500	6500	2500	0	4000	0.384615	38.46
typo50_real10	500	13000	8500	500	4500	0.653846	65.38
typo50_real40	500	13000	7000	500	6000	0.538462	53.85
typo50_real70	500	13000	4000	500	9000	0.307692	30.77
typo75_real10	500	20000	11500	3500	8500	0.575	57.5
typo75_real40	500	20000	9000	1500	11000	0.45	45.0
typo75_real70	500	20000	5000	1000	15000	0.25	25.0

Table 19: Overall external spellcheck recovery for this dataset (pooled across spellcheck result files).

dataset	configs_count	total_typo_words	restored_exact_n	changed_other_n	not_restored_n	restored_exact_frac	restored_exact_pct
gsm8k	9	118500	55000	9000	63500	0.464135	46.41